Which Exposure Matters? Predicate, Pair, and Rule-Level Exposure in Conditional Diffusion, and Optimizing the Exposure Budget

Research proposal, revision 4. Status: design stage, no experiments run. Items tagged [verified] were checked against the publisher or arXiv page (abstract level only); everything else must be checked before submission.

1. Problem statement

Conditional diffusion models fail unevenly on rare or complex scene conditions. Recent controlled work reports that scene complexity matters more than concept imbalance in low-data regimes and that compositional generalization collapses as more combinations are held out (MOSAIC, Jeong et al., ICML 2026 [verified]); that compositional abilities emerge multiplicatively from constituent abilities (Okawa et al., NeurIPS 2023 [verified]); that gradient conflicts between head and tail classes may drive long-tail failures (GRASP, Nützel et al., arXiv 2025 [verified], a fine-tuning study on pretrained models); and that local conditional scores underlie compositional generalization (Bradley, ICML 2026 [verified]).

What I could not find in a limited search is a quantitative account of *which level of exposure a rule needs*. A rule such as “a red object left of a striped square” can be rare because (a) its predicates are rare, (b) its predicates are common but almost never co-occur, or (c) only the full conjunction is rare while its pairs are common. These are different training signals, and class frequency conflates them. A PMI-based analysis separating pair co-occurrence from single-concept frequency exists for vision-language pretraining (arXiv 2507.08000 [verified, abstract level]); I found no equivalent controlled study for diffusion training.

Central problem. For a target rule r with predicate set P_r, how does the probability that a trained conditional diffusion model satisfies r depend on exposure at the predicate, pair, and rule level; how does that dependence change with relational complexity; and can a fixed sampling budget be allocated across rules to maximize worst-case rule satisfaction?

2. Positioning: what is and is not claimed

Not claimed as novel: long-tailed diffusion training, synthetic compositional benchmarks, timestep-resolved loss analysis, or the existence of gradient conflict in diffusion training (MOSAIC, Okawa, Go et al., GRASP all precede this work).

Candidate contributions, each to be defended against the closest prior work:

1. **Marginal-preserving exposure manipulation.** Rule-level exposure is varied while predicate marginals are held fixed, with *lift* (log ratio of observed to independence-predicted joint exposure) as the dose variable.
2. **An independence baseline and composition gap.** A falsifiable null built on Okawa et al.'s multiplicative finding, with the deviation from it as the primary quantity.
3. **Exposure thresholds (N50)** rather than accuracy at one budget, which avoids floor and ceiling effects.
4. **Interference measures with causal validation and an incremental-validity test**, rather than correlational gradient statistics.
5. **Exposure allocation as a constrained optimization problem**, gated on the surrogate model being predictive.

Overlap risk is highest with MOSAIC. Before any claim, read its full text and run its released code (listed on the PMLR page as github.com/eugene6923/MOSAIC).

3. Research questions and falsifiable hypotheses

ID	Hypothesis	Falsified if
H1	Composition costs: observed rule satisfaction falls below the independence baseline (composition gap $\Gamma < 0$), more so as relational complexity K_{rel} rises	Γ lies inside the equivalence margin δ in all K_{rel} cells
H2	At fixed predicate marginals, Γ increases with lift λ; the slope is larger for relational than for attribute-only rules	The slope's CI lies inside $\pm\delta$, or slopes do not differ by K_{rel}

ID	Hypothesis	Falsified if
H3	Zero-shot success on a held-out rule requires pairwise support: leaving out only the rule is survivable, leaving out its pairs is not	Zero-shot success is equal across the two leave-out types
H4	Breadth matters beyond frequency: predicates seen in many partner contexts yield lower N50 than equally frequent predicates seen in few	N50 shows no dependence on breadth after controlling for frequency
H5	Update-transfer features improve out-of-run prediction of per-rule learning speed beyond exposure and complexity, and their sign predicts multi-step exposure-ablation effects	No cross-validated gain, or sign agreement is at chance
H6	Composition-gap loss is concentrated in specific log-SNR bands, predictable from the dataset's per-frequency SNR, and for relational rules sits in inter-object gap and boundary regions	Excess loss is spread uniformly over bands and image regions

The optimization contribution (Section 10) is conditional on H2 or H5 holding.

4. Formal framework

4.1 Exposure hierarchy

A scene s has ground-truth annotations. A rule r is a conjunction of predicates P_r over existentially quantified, distinct object variables (unary attribute predicates and binary relations). $s \models r$ if some assignment of distinct objects satisfies every predicate. Let π be the training distribution over scenes. For any predicate set S define the exposure $n(S) = \Pr_{s\sim\pi}[s \models S]$, computed from realized ground-truth annotations, not from which template generated the scene. This matters because a scene generated for one rule also satisfies every weaker rule it contains.

- Predicate marginal: $m_p = n(\{p\})$.
- Rule exposure: $n_r = n(P_r)$.
- Independence reference: $n_r^{ind} = \prod_{p\in P_r} m_p$.

- **Lift:** $\lambda_r = \log(n_r / n_r^{ind})$. Negative lift is a rare combination of common primitives.
- Pairwise lift: $\lambda_{pq} = \log(n(\{p,q\}) / (m_p m_q))$; λ_r^{min} is the minimum over pairs in P_r.
- **Breadth:** $b_p = \exp(H_p)$, where H_p is the entropy of the distribution of co-occurring predicates among scenes satisfying p. It separates frequent-but-narrow from frequent-and-broad predicates, which is the useful meaning of "sharing".

Total exposure over training is $N_{seen} \times n(S)$, with N_{seen} = steps × batch size.

4.2 Independence baseline and composition gap

Let c_p be the rule-satisfaction rate (RSR) of the single-predicate rule {p}. The null is $\hat{c}_r = \Pi_{p \in P_r} c_p$. The composition gap is $\Gamma_r = \text{logit}(RSR_r) - \text{logit}(\hat{c}_r)$. The product null is a heuristic inspired by Okawa et al., so a log-additive and a min-competence null will be reported as sensitivity analyses.

4.3 Dose–response and threshold

Across runs each rule receives different exposure. Fit $\text{logit } RSR_r = a_r + \beta_r \log n_r$ (hierarchically across rules) and define $N50_r$ as the exposure at 50% RSR. Rule-level N50 is the main outcome for complexity and breadth effects.

5. Benchmark

Renderer. 64×64 images, three objects per scene, anti-aliased, with enforced minimum separation and visibility. Vocabulary: 3 shapes × 4 colours, with texture (plain, striped) included only if a pilot shows it is legible at this scale; otherwise drop it or move to 96×96. Relations: left-of, above, plus one further relation after the pilot. Position is a relation, not an attribute.

Rule grammar. K_{attr} counts unary predicates and K_{rel} counts relations, over object variables that may be unconstrained ("a red object left of some object"). Cells are feasible as follows: $K_{rel} = 0$ needs 1–3 attributed objects; $K_{rel} = 1$ needs two variables; $K_{rel} = 2$ needs three variables in a chain or two relations on one pair. The pilot builds the ($K_{attr} \in \{1,2,3\}$) × ($K_{rel} \in \{0,1,2\}$) table, flags infeasible cells, and reports object count and foreground area as covariates. Unspecified objects are distractors with randomized attributes; their incidental exposure is counted through realized annotations. Rules need not be mutually exclusive.

Templates. About 54 training templates across the feasible cells, plus a held-out set for the leave-out probes (H3). Concrete shapes, colours, and relations are counterbalanced across complexity levels.

Checker. A classical, deterministic pipeline: colour and texture masks, connected components, centroids, geometric relation tests, with mask-based handling of touching or overlapping objects. It is validated on clean renders, perturbed renders, and a human-audited, complexity-stratified sample of generated images from a pilot model (at least 300). False-accept and false-reject rates are reported per predicate and per rule, with a target of 2% or less on real renders.

6. Controlling exposure with a linear program

Let Q_r be the scene distribution rendered for template r, and let the training distribution be $\pi = \Sigma_r w_r Q_r$ with w on the simplex. For any predicate set S, $n(S) = \Sigma_r w_r A_{S,r}$, where $A_{S,r} = \Pr_{s\sim Q_r}[s \models S]$ is estimated once by Monte Carlo (about 10^5 scenes per template). Exposure statistics are therefore linear in w.

A design is a target for predicate marginals m^* and for lifts on chosen rules. Fixing marginals constrains each $n(P_r)$ linearly, so feasibility is a linear program, and a design closest to uniform solves $\min \mathrm{KL}(w \,\|\, \text{uniform})$ subject to the linear constraints. Fixing roughly one marginal per predicate leaves a subspace of free directions in which lifts can vary. Stage 0 computes that subspace, samples designs from the feasible polytope, and reports the achieved correlations and variance inflation factors among log marginal, lift, breadth, K_{attr}, and K_{rel}. The experiment proceeds only if lift is decorrelated from marginal exposure ($|r| \leqslant 0.3$, VIF < 2).

Leave-out probes (H3) are special cases: set $n_r = 0$ (rule removed), or additionally zero the pair exposures within P_r (pairs removed).

7. Model and training

Online sampling is primary. Scenes are rendered fresh each step, so frequency means sampling probability, and memorization and repeated exposure are removed as confounds. A secondary finite-dataset regime (fixed N_{tot}, repeated epochs) links the results to MOSAIC's dataset-size findings and to the memorization phase reported by Okawa et al.

Model. A small conditional U-Net trained from scratch, with fixed noise schedule, AdamW, EMA, 10% condition dropout, fixed sampler, and a guidance-scale sweep ($w = 1$ plus one locked value). Budget and capacity are calibrated in the pilot so that median reference RSR lies roughly in [0.3, 0.7]. A second capacity is run later.

Conditioning. Primary: a structured scene-graph encoder (predicate tokens with object-slot identifiers, cross-attention). Secondary arms in Stage 4: categorical rule ID (seen rules only), factorized attributes without relations, and relation-aware, with matched parameter counts. Arm B cannot represent relation-only differences, so affected rules are flagged.

8. Outcomes and image-processing diagnostics

- **Primary:** RSR per rule and run; N50; composition gap Γ.
- **Loss:** excess loss against a reference model trained with uniform π at the same budget, on identical held-out scenes, noise, and timesteps, in five log-SNR bands. Null-conditioned loss is reported separately, because under imbalance the null model learns the training marginal. A wrong-condition loss measures how class-specific the conditioning is.
- **Region-resolved error.** Using renderer masks, the error is split into object interior, boundary ring, inter-object gap, and background, normalized by area. Relations live in the gaps and boundaries, so this is where H6 predicts relational deficits.
- **Spectral analysis.** For band f, compute the analytic noise level at which the dataset's power in f equals noise power. This gives a predicted map from log-SNR to spatial frequency, which is tested against where composition-gap loss concentrates. Errors are normalized by each rule's own spectrum, since texture adds high-frequency energy. Phase and layout are handled by the mask analysis, since magnitude spectra do not encode relations.
- **Diversity and memorization:** valid-configuration coverage and nearest-neighbour distance to training scenes, interpreted jointly.
- **Trajectories:** checkpointed per-rule loss and RSR, and time-to-threshold, to separate slower learning from a lower asymptote.

9. Optimization analysis

Counterfactual transfer. From a checkpoint θ with optimizer state ω, clone state, apply K AdamW steps ($K \in \{1, 10, 100\}$) using batches from rule q, and measure the change in rule r's loss on a fixed held-out set with common random numbers. This yields $T_{r \leftarrow q}$. Controls: self-transfer from fresh batches of rule r, and a random-condition batch. Report T normalized by $|T_{r \leftarrow r}|$, per U-Net block and per SNR band. A single step has a tiny first-order effect, so the multi-step variants carry the weight.

Gradient geometry. Cosine similarity with per-layer computation and a within-rule split-half noise floor. Opposing gradients are not treated as harmful interference without the transfer evidence above.

Causal exposure ablation. From a shared checkpoint, continue training for M steps with q's sampling mass removed or doubled (renormalized), against an unmodified continuation with the same random seed, and measure the effect on r. This tests whether the sign of T predicts multi-step effects.

Incremental validity (H5). Compare a base model of per-rule learning speed ($\log n_r$, λ_r, mean marginal, breadth, K_{attr}, K_{rel}) with the same model plus transfer features, using leave-one-run-out cross-validation. The mechanism claim is made only if held-out deviance improves by a prespecified margin with a CI excluding zero.

Ablations and baselines. Weight-decay exclusion on conditioning parameters; limited SGD versus AdamW; Min-SNR weighting; class-balanced resampling; CBDM-style regularization if time permits.

10. Gated optimization contribution: exposure allocation

If H2 or H5 holds, fit a surrogate for per-rule success as a function of the allocation:

$\text{logit}\ \hat{p}_r(w) = a_r + \beta_r \log n_r(w) + \gamma\, \lambda_r(w) + \delta \cdot \log \bar{m}_r(w) + \varepsilon \cdot \log \bar{b}_r(w)$ [+ sparse transfer terms from Section 9].

Solve max over w in the simplex of a softmin (log-sum-exp smoothing) of $\hat{p}_r(w)$, or of the mean of the lowest decile, with a support floor $w_r \geqslant w_{min}$. Since n and m are linear in w, the problem is smooth and low-dimensional; use exponentiated-gradient (mirror descent) with multiple restarts, and report convergence and sensitivity to the smoothing temperature. Compare against uniform-over-rules, uniform-over-predicates, inverse-frequency sampling, and loss-reweighting baselines at equal compute, on fresh seeds and held-out templates not used to fit the surrogate.

Gate: proceed only if the surrogate predicts held-out runs with mean absolute RSR error ⩽ 0.10. Otherwise report the surrogate's failure as a negative result and stop. No other intervention is tested.

11. Statistical plan

Unit of analysis. Rule (template) crossed with run. A run is one (seed, exposure design) pair, so seed and design are not confounded. Rules within a run share one model and one budget, so run enters as a random effect.

Primary model (binomial GLMM, logit link):

$$\text{logit } p_{r,k} = a_r + \beta_\lambda \lambda_{r,k} + \beta_{\lambda R} (\lambda_{r,k} \times K_{rel,r}) + \beta_m \log \tilde{m}_{r,k} + \beta_b \log \bar{b}_{r,k} + \theta \cdot (K_{attr}, K_{rel} \text{ as categorical contrasts}) + u_r + v_k + w_{rk},$$

with template, run, and template-by-run random effects, and a template-level random slope on λ if identifiable. **Primary estimands:** β_λ and $\beta_{\lambda R}$, tested with Holm correction as a family of two, against an equivalence margin δ fixed from the pilot before confirmatory runs. A three-way interaction is deliberately not primary: it is the lowest-powered target. All other analyses are exploratory with Benjamini–Hochberg control, and the registered analysis lists the exact contrasts.

Floor and ceiling. Calibrate budget in the pilot so most rules are mid-range. Exclusion rules (reference RSR below 0.05 or above 0.95) are fixed from pilot reference runs only, and excluded rules are reported separately.

Power. The pilot trains all templates jointly (cost does not grow with the number of templates), with at least three seeds, since two seeds cannot identify a variance component. Simulate the GLMM from pilot variance components, 1,000 or more simulations per candidate design, and choose the smallest design with about 80% power and informative intervals. Increase runs, not generated images, if underpowered. Generated images from one model are never treated as independent replicates; uncertainty comes from resampling runs and templates.

12. Staged plan, compute, go/no-go

Stage 0: instrument checks. Renderer, grammar table, checker validation, A-matrix, LP feasibility and covariate decorrelation, an analytic Gaussian-mixture denoiser to validate loss normalization, and a small replication of MOSAIC's complexity-versus-imbalance

contrast on this renderer. *Go if* checker error ⩽ 2%, the design polytope meets the decorrelation criterion, and the replication matches MOSAIC's direction.

Stage 1: pilot. All templates, reference plus three exposure designs, at least three seeds, short budget. Calibrate capacity and budget, estimate variance components, fix δ, run the power simulation.

Stage 2: confirmatory exposure experiment. Designs and seeds set by power analysis (working estimate: six designs × three seeds plus references), including the H3 leave-out probes.

Stage 3: diagnostics. Transfer, exposure ablation, region and spectral analyses, trajectories, incremental-validity test.

Stage 4: gated. Conditioning-arm comparison and the allocation optimization, on fresh seeds and held-out templates.

Stage 5: robustness. Second capacity; optional small CLEVR-style check for relational claims.

Compute. Working assumption, to be replaced by pilot measurements: about 3 GPU-hours per run on one modern GPU, roughly 60 runs, plus about 50 GPU-hours for diagnostics, so on the order of 200–250 GPU-hours. Set a hard cap after the pilot.

13. Risks and limitations

- **Overlap.** MOSAIC and Okawa et al. are close; the contribution rests on the marginal-preserving design and the dose–response framing, and must be confirmed against full texts.
- **Independence null.** The product baseline may be a poor null for relations; sensitivity analyses address this but cannot remove it.
- **Design entanglement.** Not all lifts are independently settable; the feasible polytope bounds what can be claimed.
- **Online versus finite data.** Online sampling removes memorization, so conclusions may differ in finite-data regimes; the secondary experiment checks this.
- **Run-level coupling.** Rules compete within one model, so rule-level effects partly reflect the whole design; the run random effect and global design covariates mitigate this.
- **Capacity regime.** Too small a model fails on everything, too large hides the effects. Calibrate, then replicate at a second capacity.

- **Checker validity.** Per-rule false-reject differences can masquerade as model effects.
- **External validity.** Geometric scenes may not transfer to photographs.
- **Novelty not established.** My literature check was limited to a few searches and abstract-level reads.

14. References

1. Jeong, Uselis, Laina, Oh, Rohrbach (2026). When Do Diffusion Models Learn to Generate Multiple Objects? ICML 2026, PMLR 306:51250–51282. https://proceedings.mlr.press/v306/jeong26b.html [verified]
2. Bradley (2026). Local Mechanisms of Compositional Generalization in Conditional Diffusion. ICML 2026, PMLR 306:9531–9565. https://proceedings.mlr.press/v306/bradley26a.html [verified]
3. Okawa, Lubana, Dick, Tanaka (2023). Compositional Abilities Emerge Multiplicatively. NeurIPS 2023. arXiv 2310.09336 [verified]
4. Nützel, Dombrowski, Kainz (2025). GRASP: Guided Residual Adapters with Sample-wise Partitioning. arXiv 2512.01675 [verified]
5. Impact of Pretraining Word Co-occurrence on Compositional Generalization in Multimodal Models. arXiv 2507.08000 [abstract-level check; authors not verified]
6. Qin et al. (2023). Class-Balancing Diffusion Models. CVPR 2023.
7. Hang et al. (2023). Efficient Diffusion Training via Min-SNR Weighting Strategy. ICCV 2023.
8. Go et al. (2023). Addressing Negative Transfer in Diffusion Models. NeurIPS 2023.
9. CORAL (long-tailed diffusion, latent entanglement), NeurIPS 2025. Authors and claims to be verified.
10. Swing-by Dynamics in Concept Learning and Compositional Generalization, arXiv 2410.08309; Compositional Generalization and Creativity in Diffusion Models, arXiv 2502.12089; Park et al. (2024) on concept-space learning dynamics. All to be read and verified.